

AWS re:Invent

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NET212-R

Application networking best practices

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Let's talk

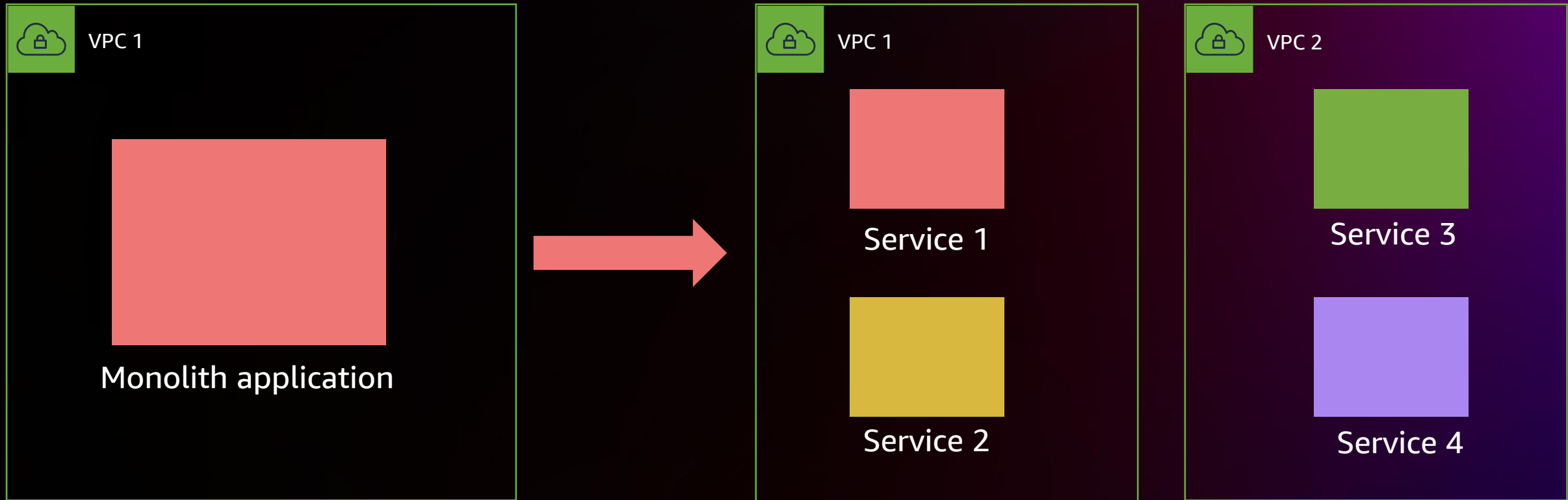
Introduction to modern application architectures

What is application networking?

What's new?

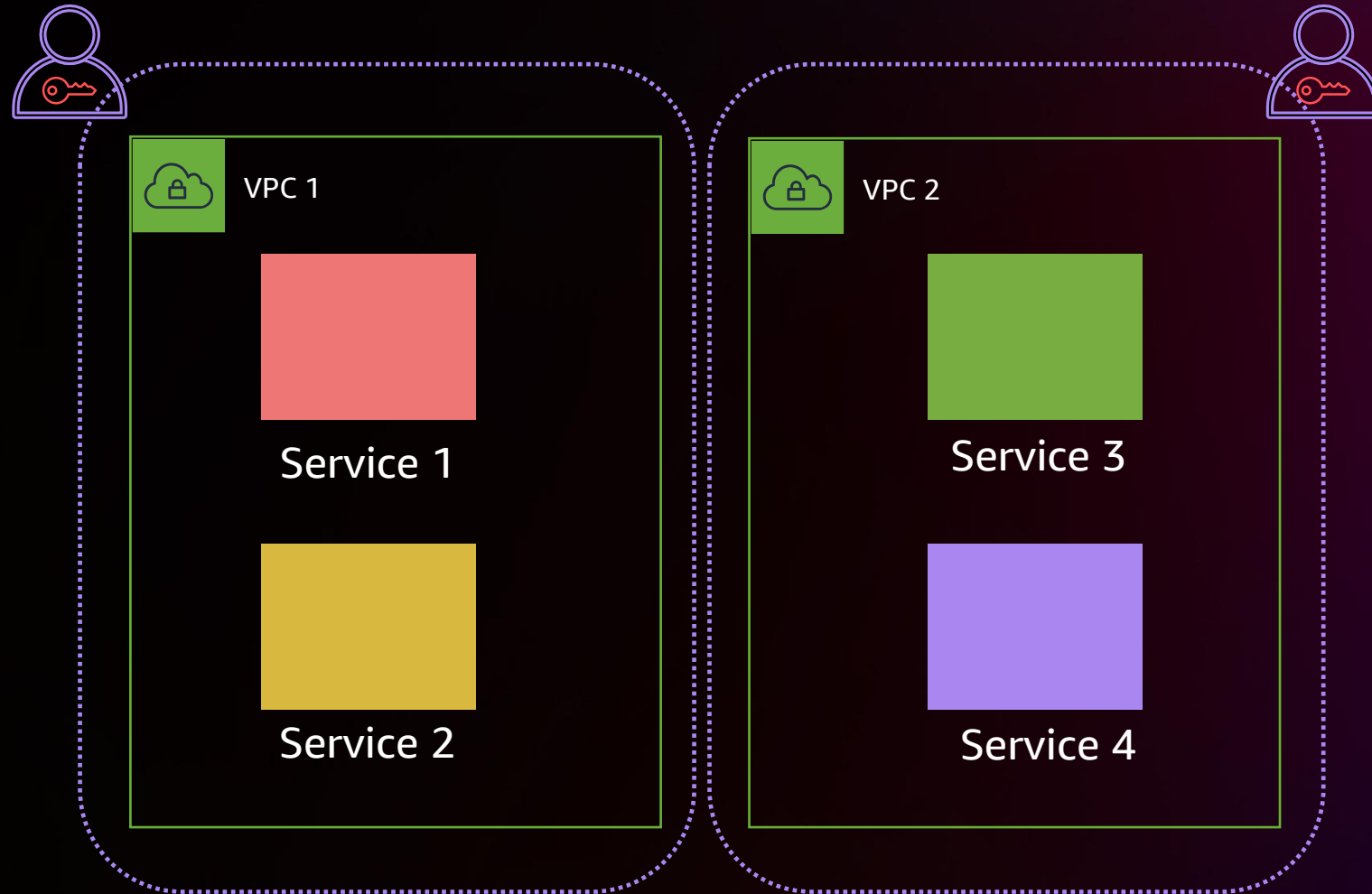
Chalk time!

The journey from monolith to microservices

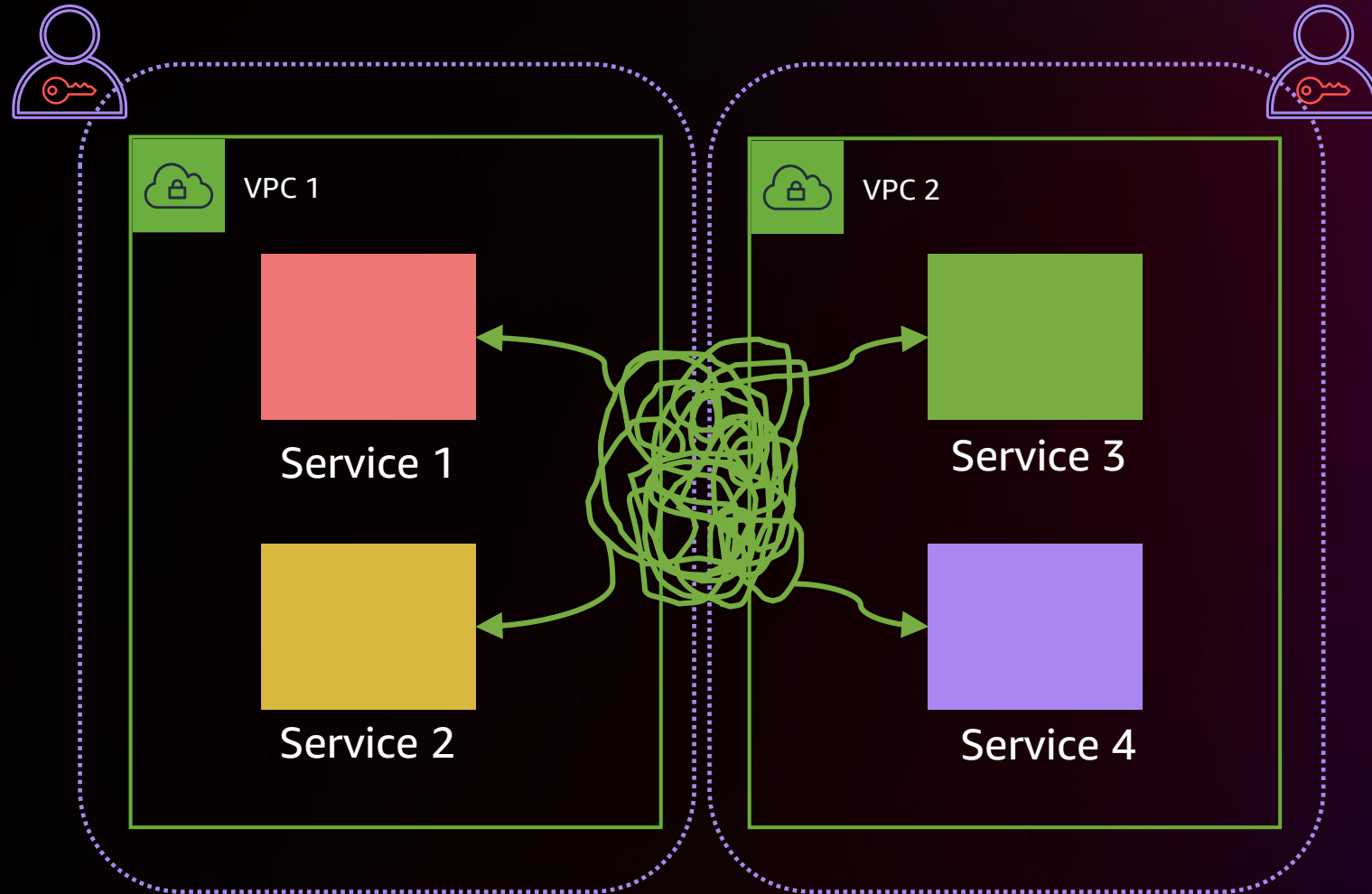


Give service teams their own sandbox

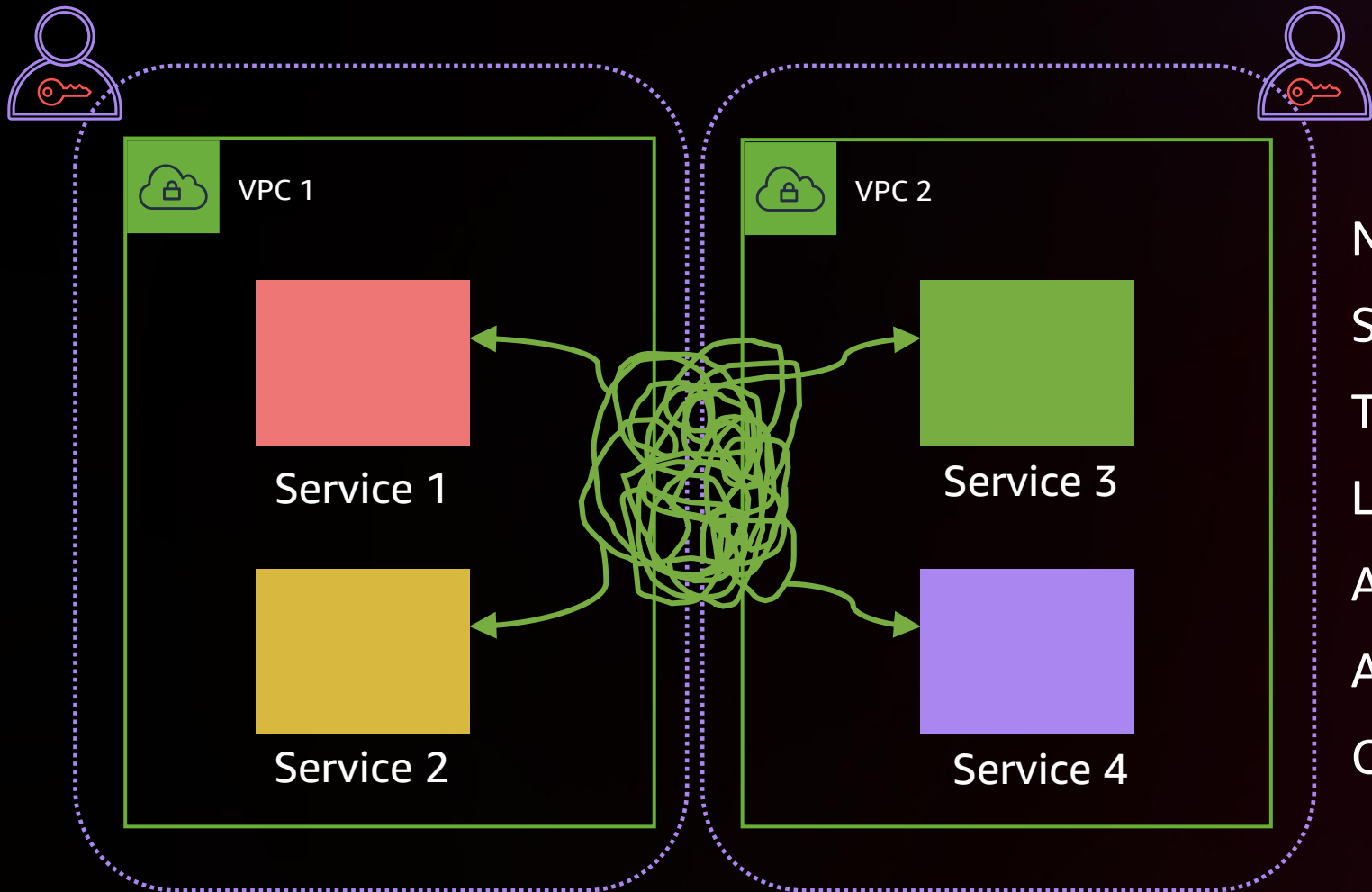
NETWORK AND PERMISSION BOUNDARIES WITH VPCS AND ACCOUNTS



But, does it match your application boundary?



How do you handle . . .



Network connectivity?

Service discovery?

Traffic management?

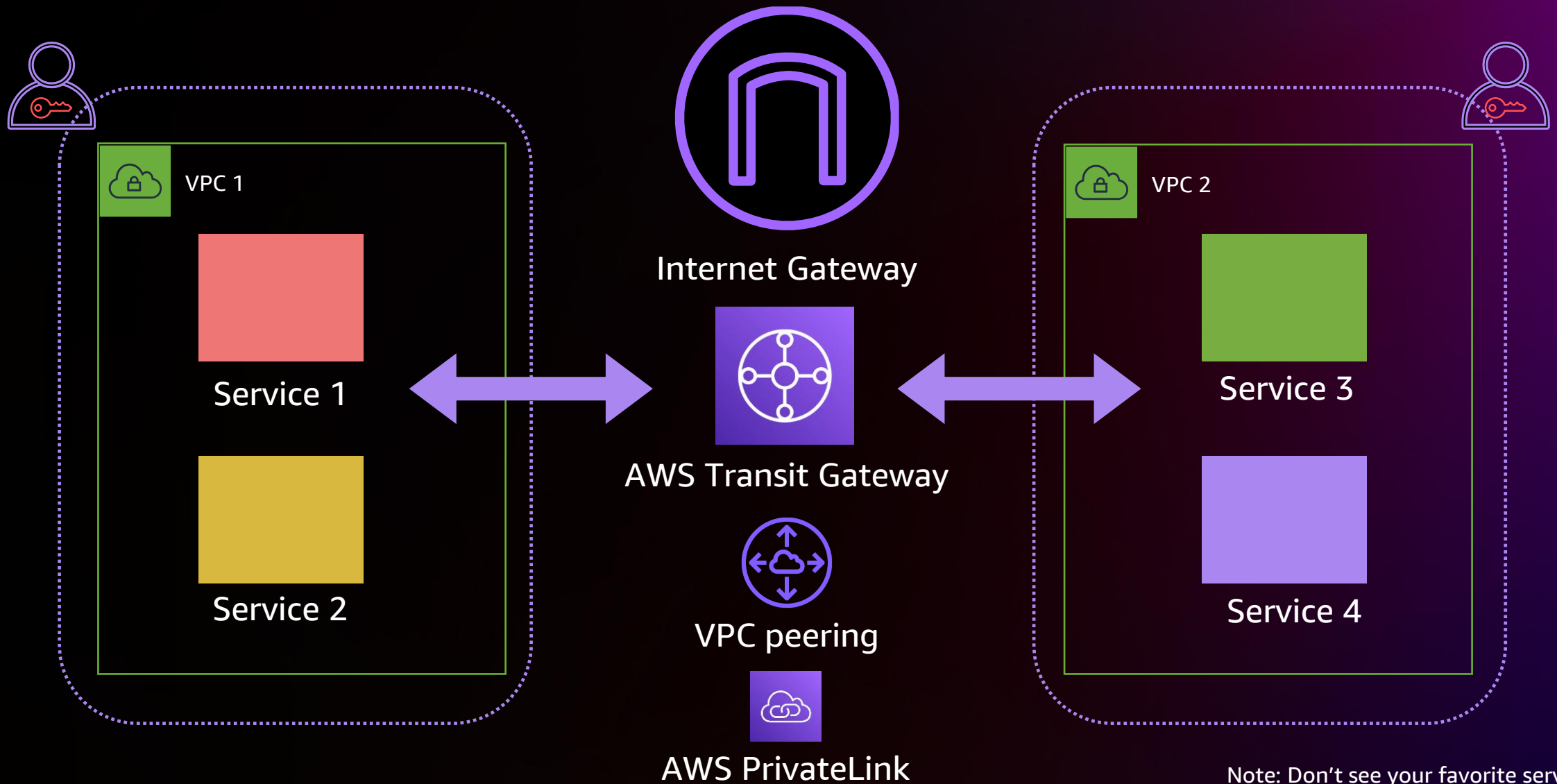
Load balancing?

Authentication?

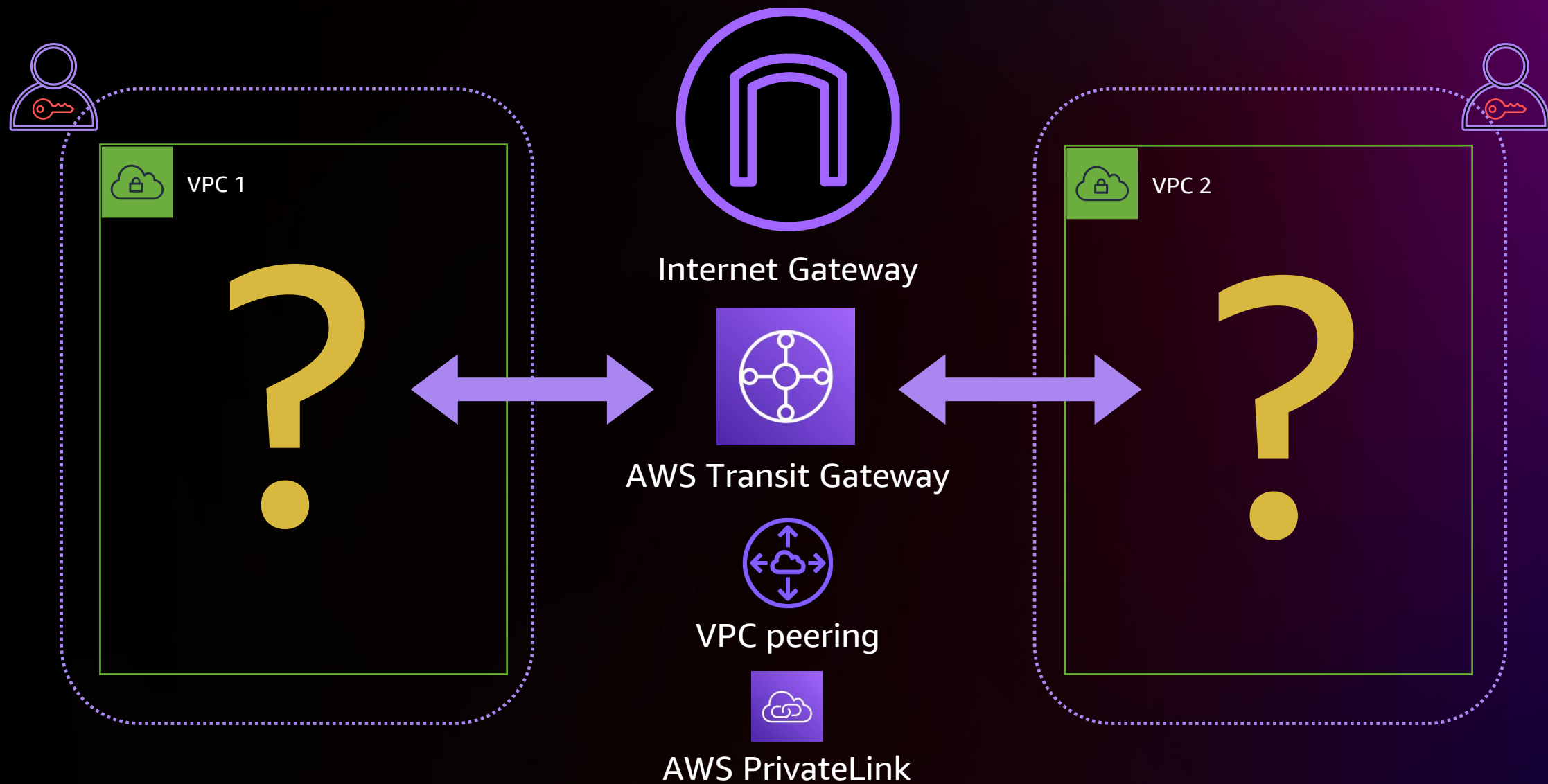
Authorization?

Observability?

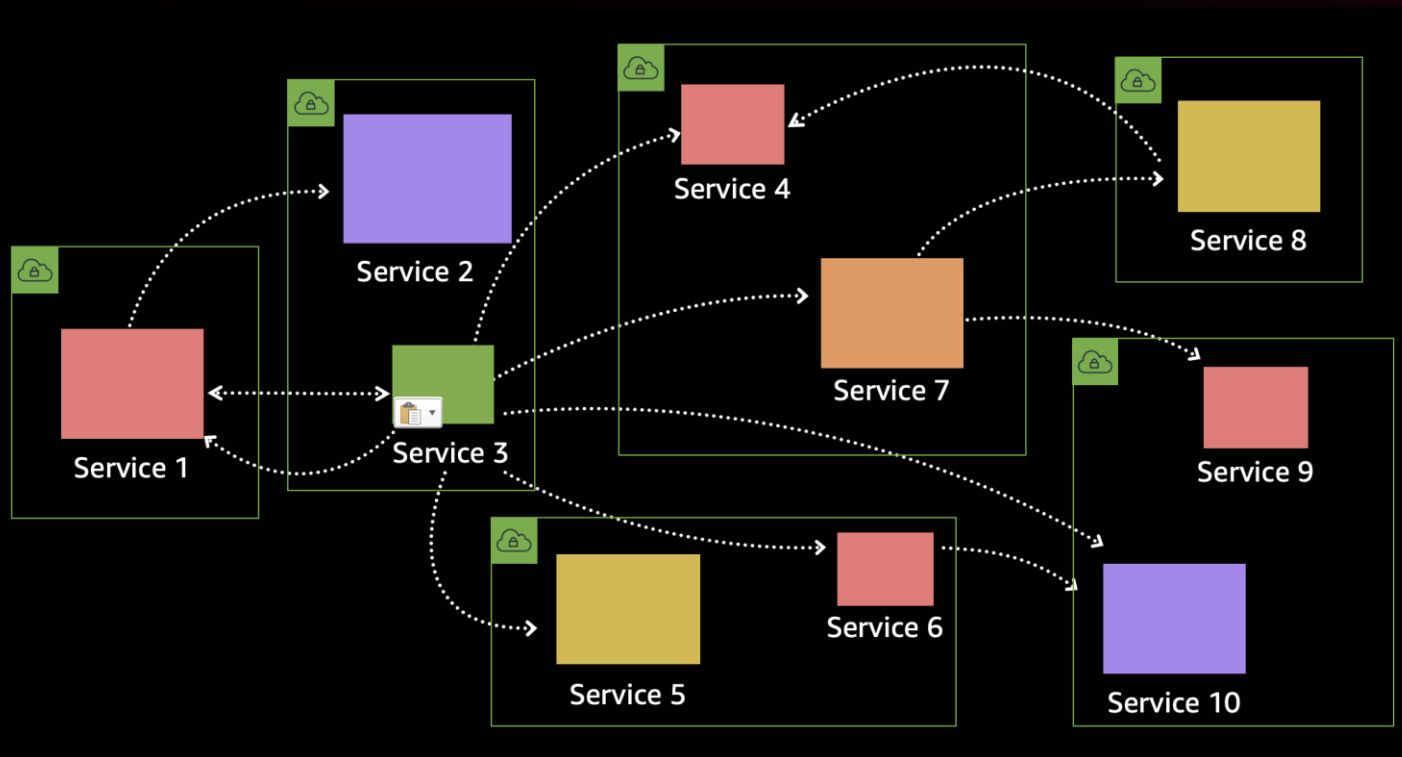
The network portion . . .



The network portion...



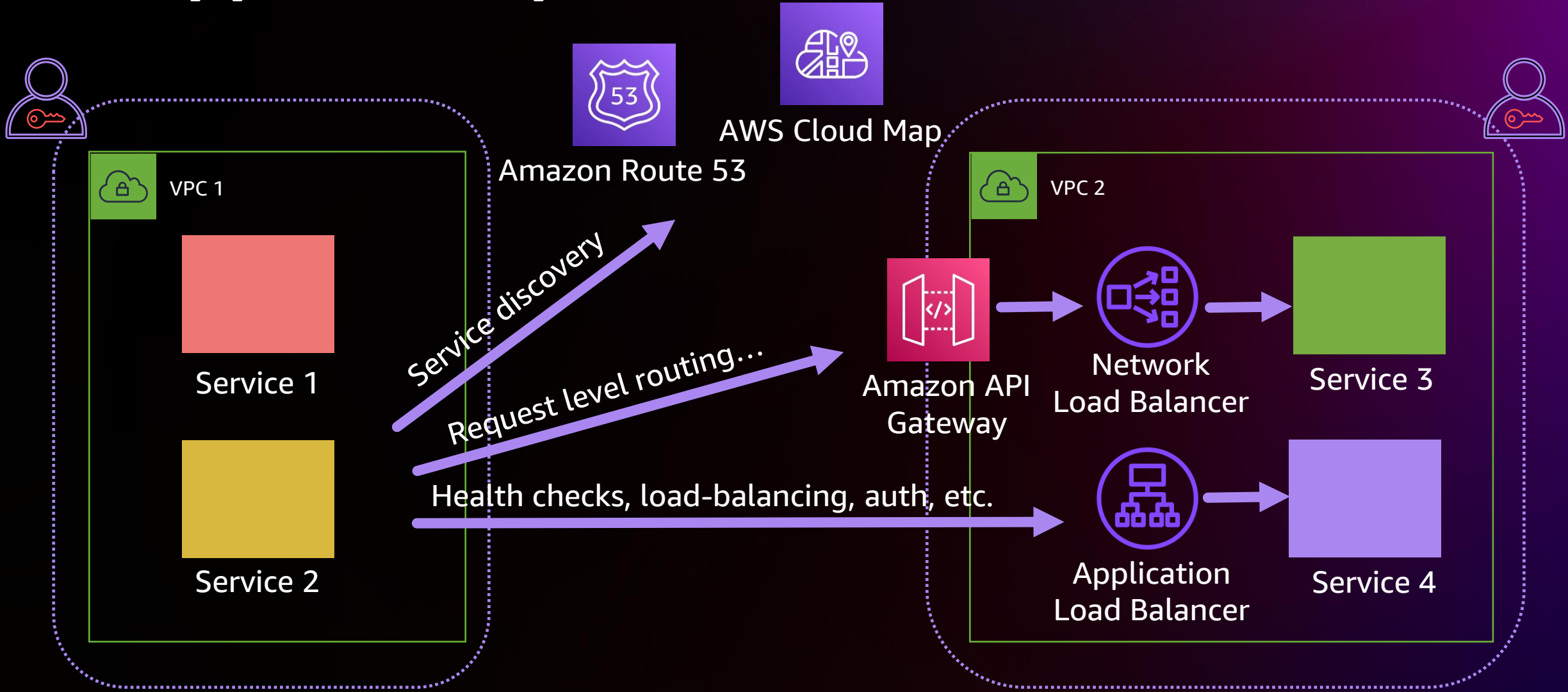
The application portion...



Service discovery
Traffic management
Load balancing
Authentication
Authorization
Observability

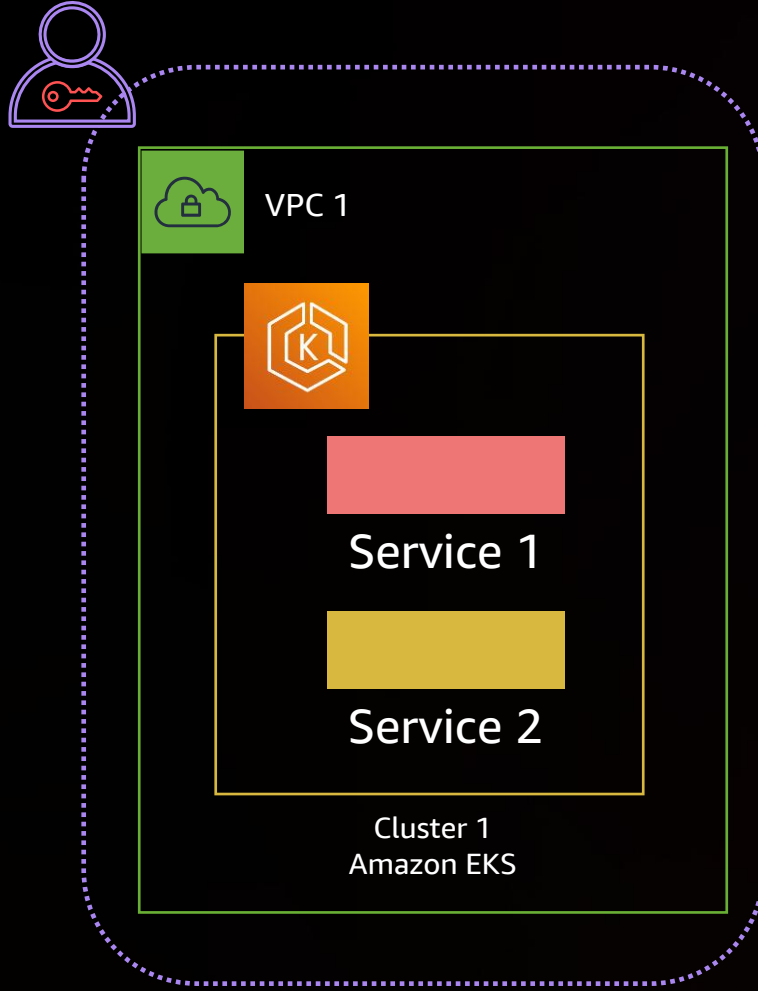


The application portion...



Containers

BASICS OUT OF THE BOX...



In VPC network connectivity ☒

Service discovery ☒

Basic round robin load balancing ☒

Traffic Management ☐

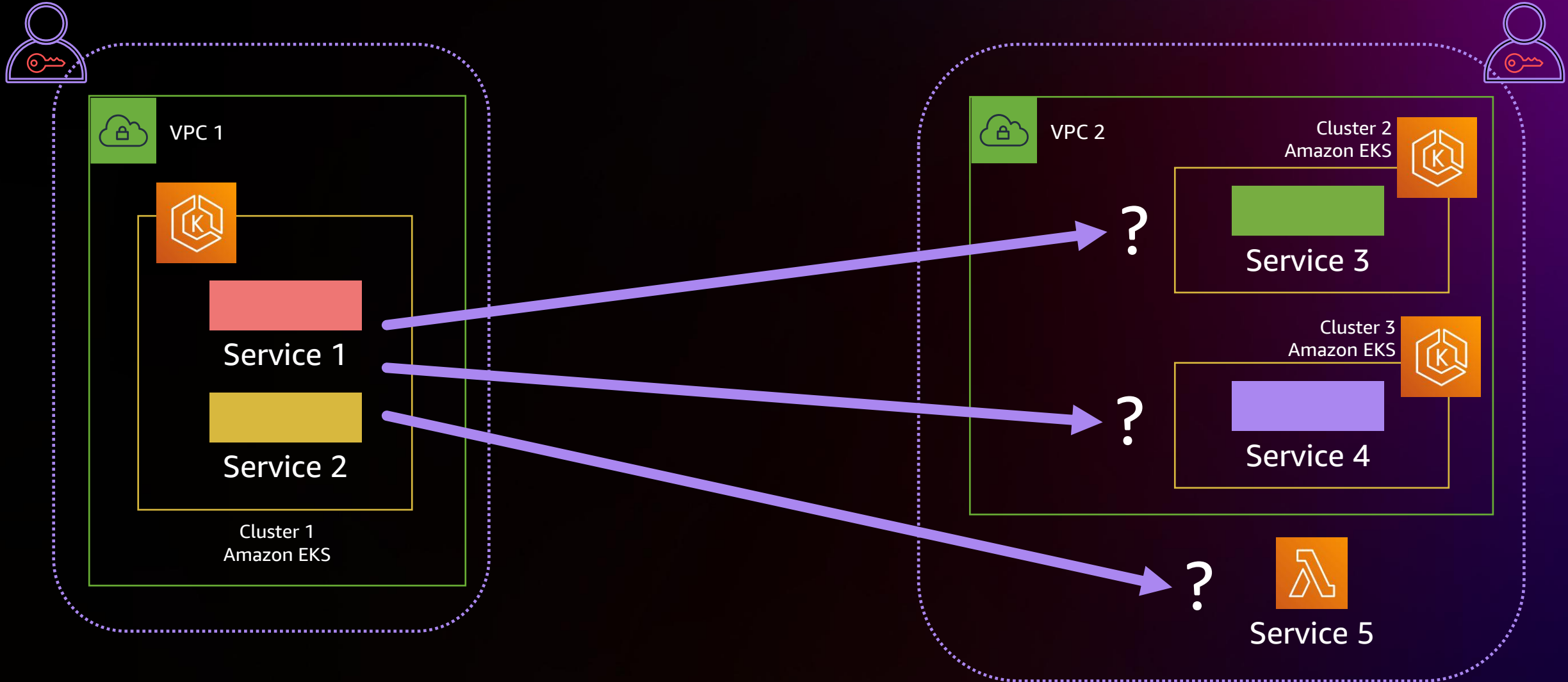
Authentication ☐

Authorization ☐

Observability ☐

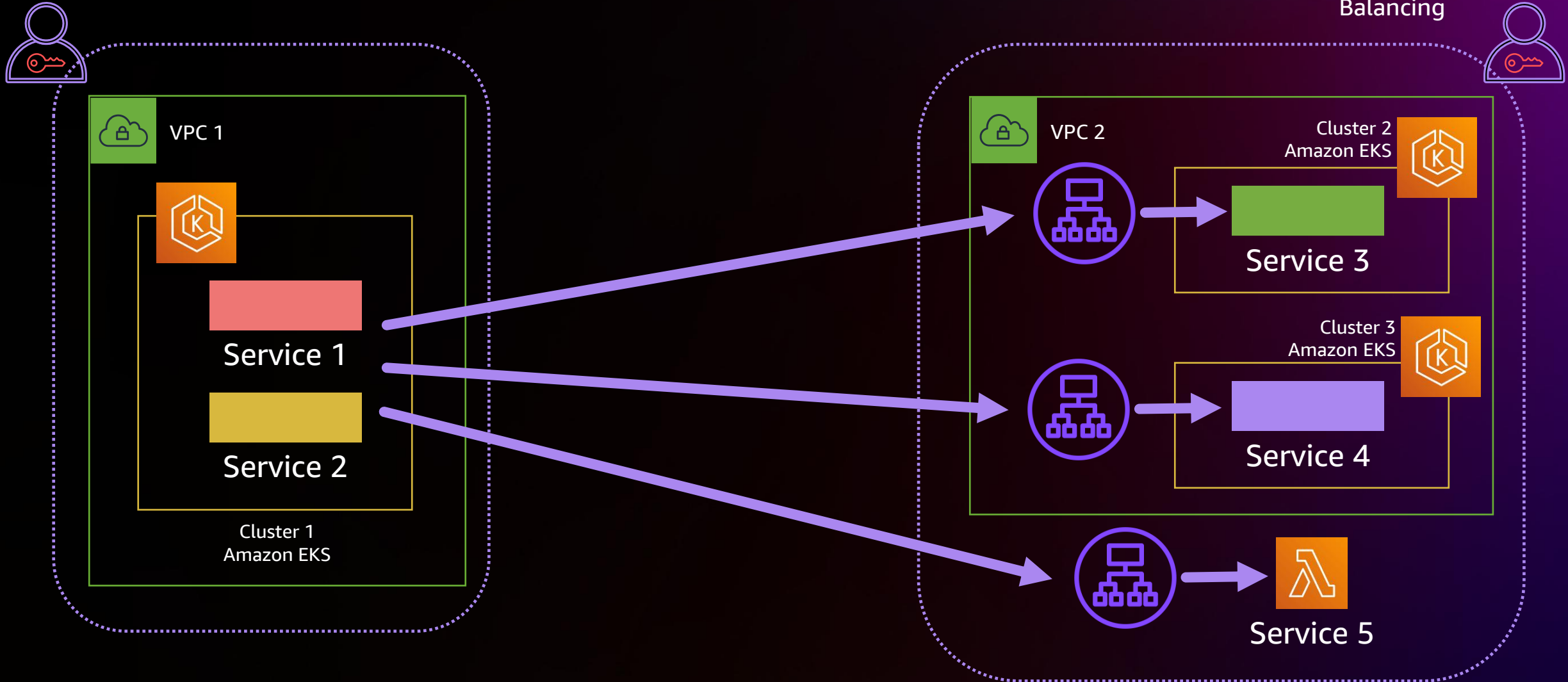
Containers

BASICS OUT OF THE BOX, BUT CLUSTER LOCAL



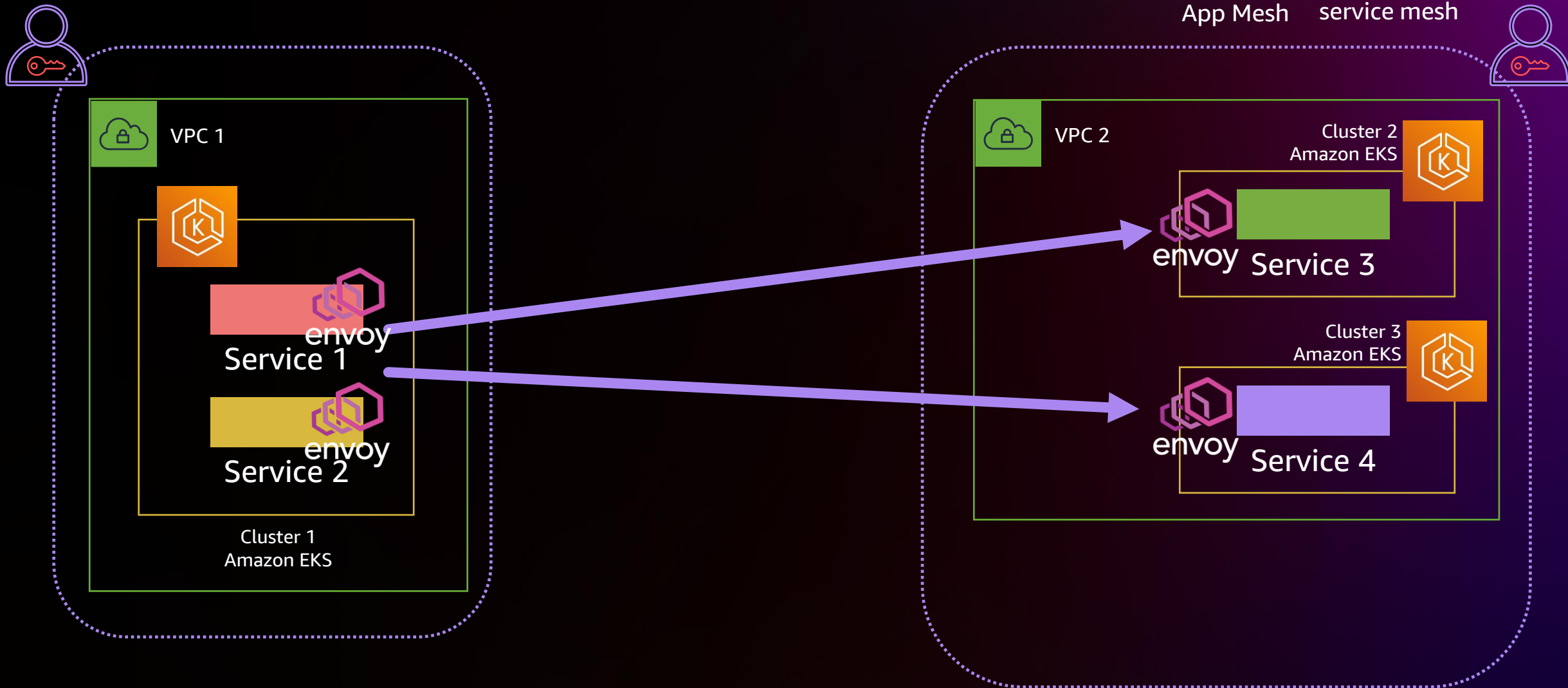
Containers

BASICS OUT OF THE BOX, BUT CLUSTER LOCAL



Containers

BASICS OUT OF THE BOX, BUT CLUSTER LOCAL

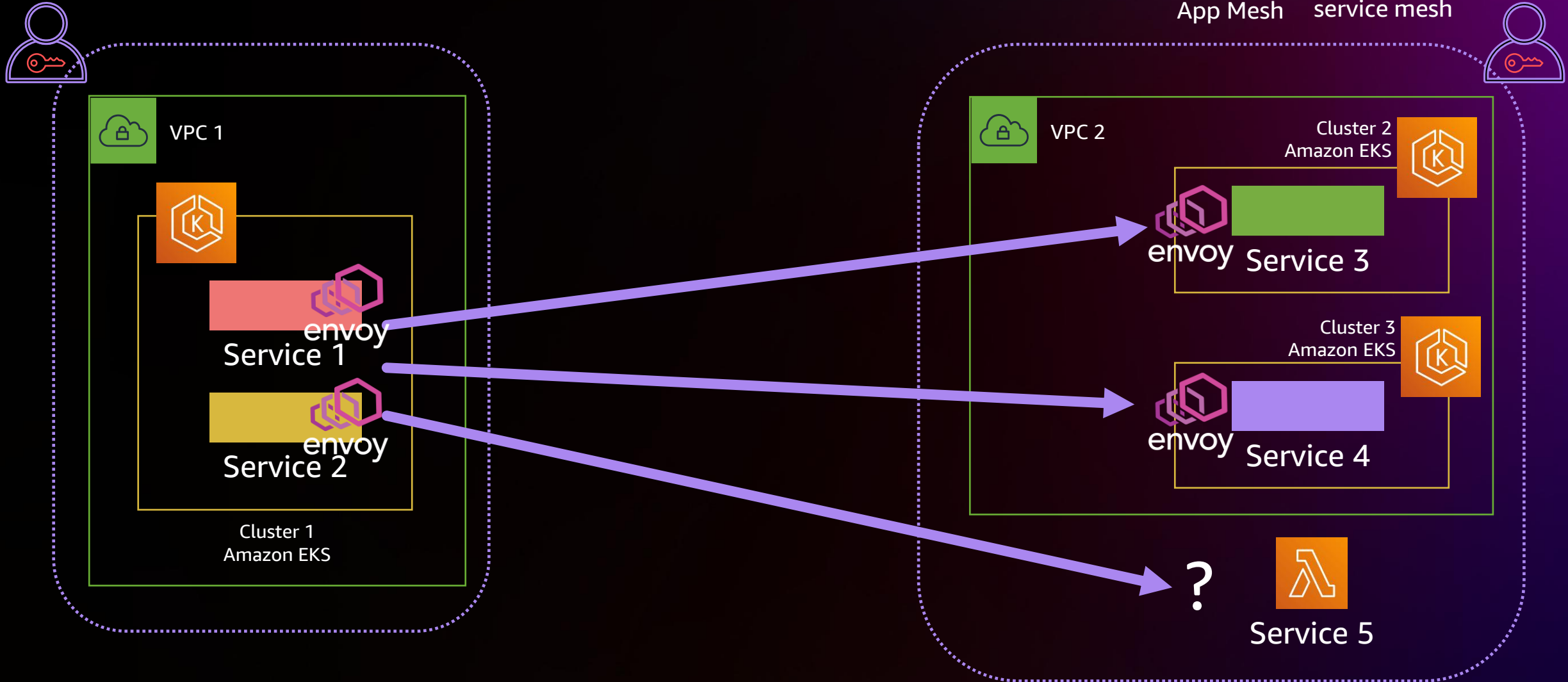


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Note: Don't see your favorite service?
Don't worry. Not an exhaustive list.

Containers

BASICS OUT OF THE BOX, BUT CLUSTER LOCAL

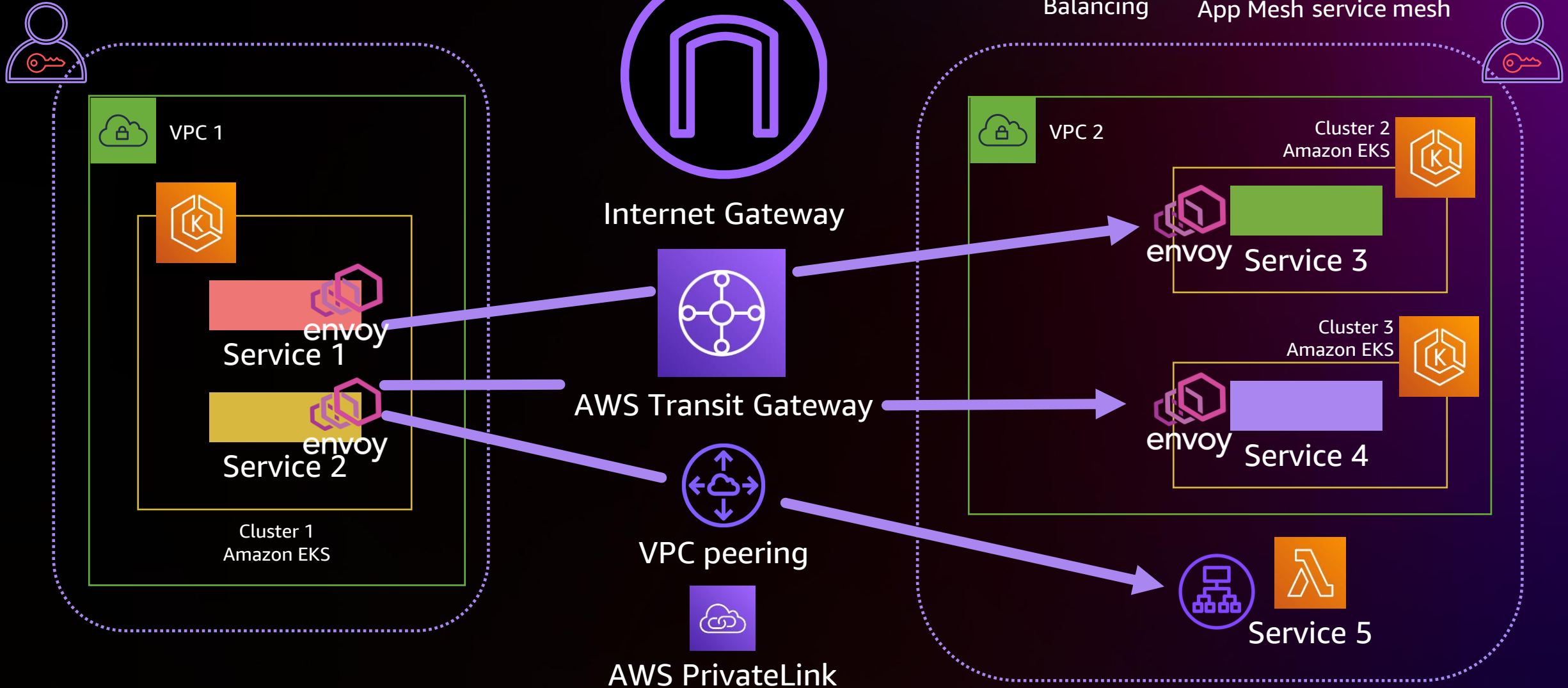


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Note: Don't see your favorite service?
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Oh yeah . . .

THE NETWORK IS STILL THERE!



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Note: Don't see your favorite service?
Don't worry. Not an exhaustive list.

Persona problem



Cloud/network admin or InfoSec

- Wants to empower developers without losing control
- Needs centralized tools to implement and enforce the organization's security posture

"I want to empower service owners to have seamless service-to-service communication, but need the ability to provide oversight, and enforce fine-grained access controls. I need observability that is integrated with my developers' systems instead of being disjointed with overlays and today's service mesh implementations."



Developer/service owner

- Doesn't want to deal with networks
- Needs advanced telemetry to help optimize and troubleshoot applications

"Service-to-service communication should just work. I want to focus on building business logic into my applications, not learning the – what's it called – OSI model?"

NEW

Amazon VPC Lattice

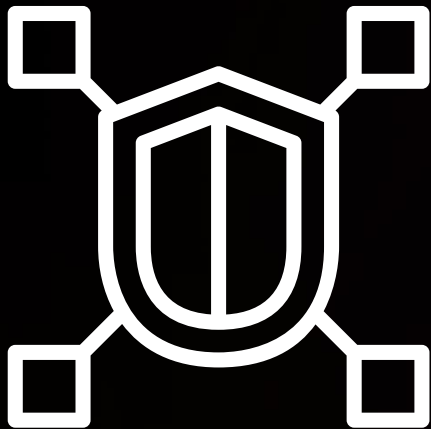
Bridging the gap between admins and developers

IN PREVIEW



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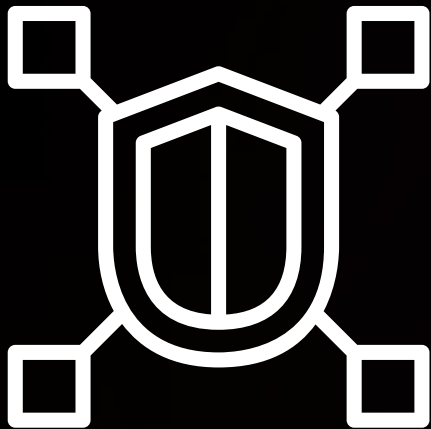
Amazon VPC Lattice



Combines network connectivity with an application layer proxy

- Automatically handles connectivity between VPCs and accounts
- Apply rich traffic control, such as application layer load-balancing and weighted targets to support blue-green deployments

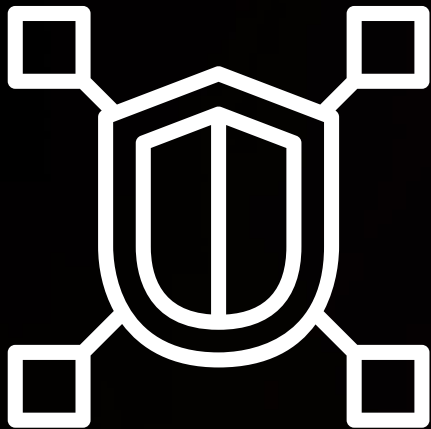
Amazon VPC Lattice



Helps implement Zero Trust principles

- Integrated with AWS Identity and Access Management (IAM): Enforce authentication and implement fine-grained authorization for your own service-to-service communication

Amazon VPC Lattice

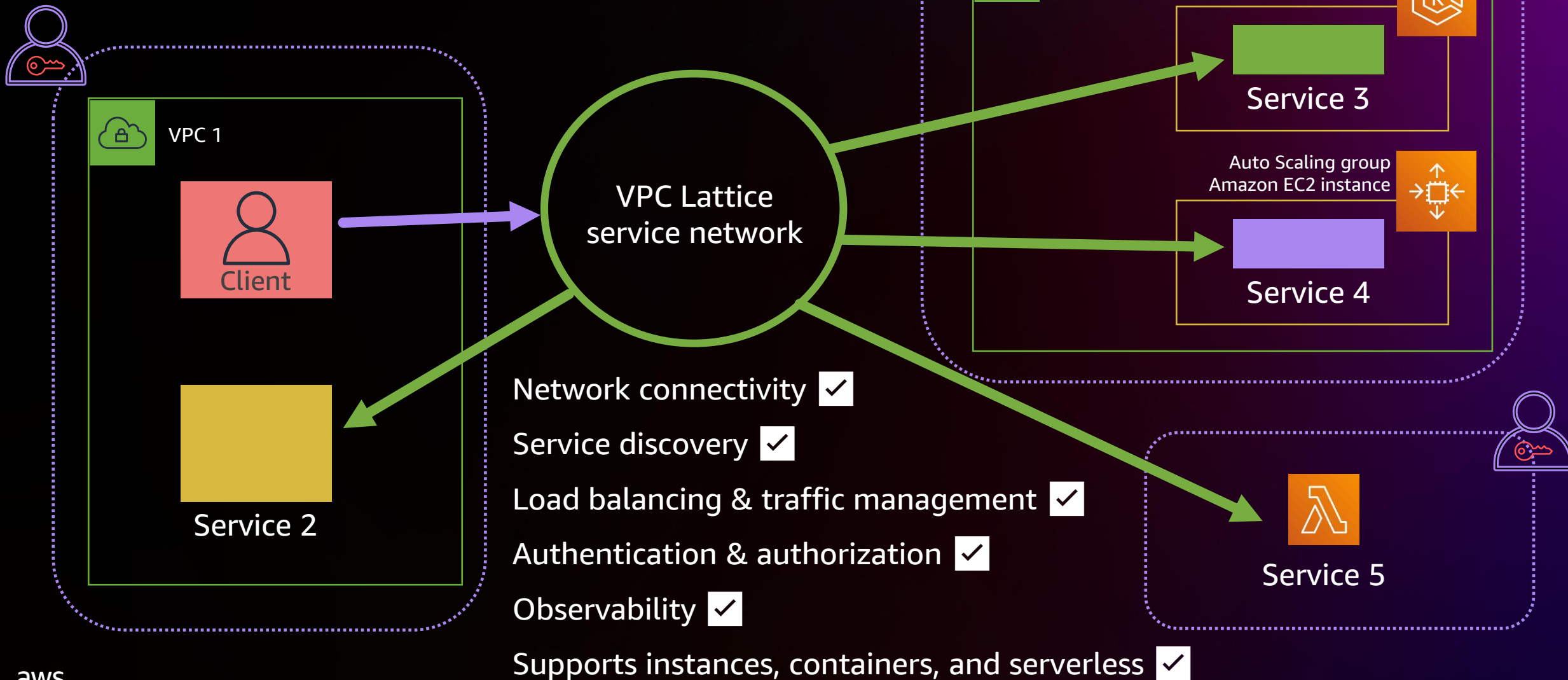


Consistent experience across instances, containers, and serverless

- Build directly into VPC and natively integrated with AWS services so that you can mix and match your compute types to meet the needs of each service

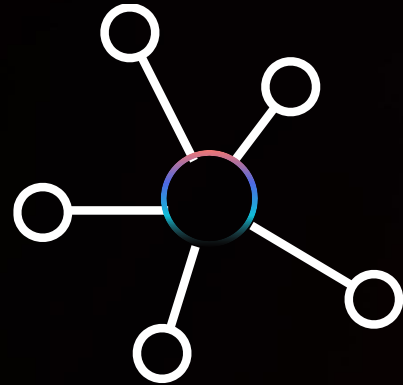
Amazon VPC Lattice

COMBINING NETWORK CONNECTIVITY AND
APPLICATION LAYER ROUTING

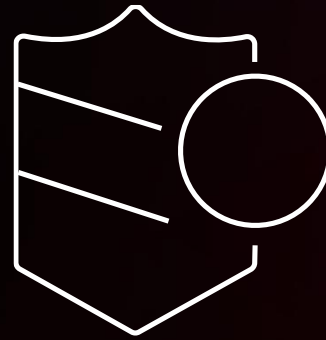




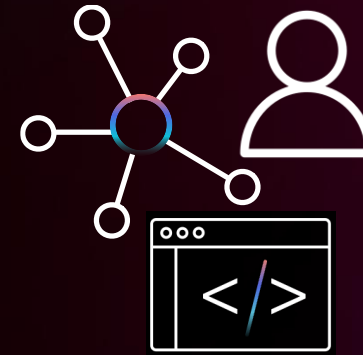
ADMIN



Create a
service network



Define access and
monitoring



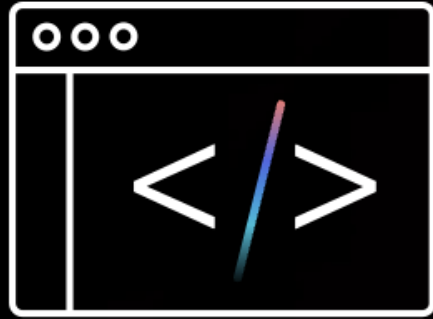
Associate
VPCs and services



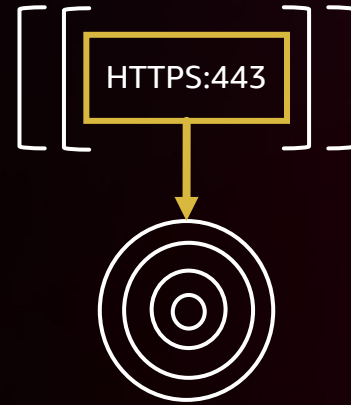
Share with other
accounts



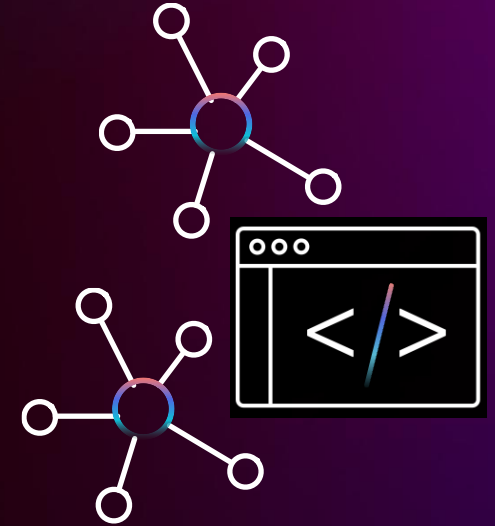
Developer



Create a service



Define routing



Associate to service
networks

Something for everyone



Cloud/network admin or InfoSec

- Can empower developers without losing control
- Enforce security guardrails
- Centralized tools to monitor service-to-service interactions



Developer/service owner

- Simple onboarding – no networking needed
- Compute flexibility to use instances, containers, or serverless
- Request level routing and load balancing to support advanced deployment patterns (blue/green, canary, etc.)
- Fine-grained and context-specific authorization
- Detailed metrics and access logs for greater visibility

Chalk time!



Thank you!

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